

# KRISHKUMAR KALYA

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## EDUCATION

### MIT ADT University

Pune, India

Bachelor of Technology in Computer Science and Engineering / **CGPA: 8.68**

Expected May 2027

**Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems, Computer Networks, Software Engineering

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL, JavaScript

**Databases:** PostgreSQL, PostGIS, MySQL, Redis

**Frameworks & Libraries:** Spring Boot, FastAPI, Flask, Spring Data JPA, Scikit-learn

**Tools & Platforms:** Git, GitHub, Docker, Postman, Linux, Maven

## EXPERIENCE

### AI-ML Virtual Internship

April – June 2025

AICTE – EduSkills, supported by Google for Developers (India Edu Program)

Remote

- Completed the 10-week AICTE–EduSkills AI-ML virtual internship, supported by Google for Developers, for fifth-semester internship credit.
- Covered TensorFlow, image classification, object detection and product image search through Google Developers ML pathways, earning a skill badge for each.

## PROJECTS

### Supply Chain Dispute Ledger | 🐙 GitHub | *Hyperledger Fabric, Node.js, Express, gRPC, JWT*

- Built a dispute resolution system on Hyperledger Fabric: supply chain partners raise disputes and attach evidence, arbiters resolve them, and every action is timestamped on an append-only ledger.
- Enforced role-based access control on-chain in the chaincode rather than at the API edge, so a compromised gateway cannot bypass authorization.
- Wrote an Express REST gateway that talks to the Fabric Gateway over gRPC/TLS, secured with JWT, Helmet and rate limiting.

### GeoAgent: Geospatial LLM Assistant | 🐙 GitHub | *Pydantic-AI, FastAPI, GeoPandas, Redis, Docker*

- Built a single async FastAPI service that maps Python geospatial functions into LLM tools through Pydantic-AI, so every model-issued call is validated against a typed contract instead of a hand-rolled planner loop.
- Implemented the spatial toolset over OpenStreetMap boundaries and Planetary Computer Sentinel-2 imagery, with geometry repair for malformed coordinates and invalid polygons.
- Streamed task logs over WebSockets from a Redis-backed state manager that replays missed updates to late-connecting clients; served Gemma via Docker Compose.

### Traffic Violation Management System | 🐙 GitHub | *Java, Spring Boot, JPA, PostgreSQL, Docker*

- Built a rule engine in Java and Spring Boot that evaluates speed events against configurable thresholds and calculates tiered fines, exempting emergency vehicles before any penalty applies.
- Designed the REST API and a Thymeleaf dashboard over PostgreSQL, with H2 for local runs.
- Tested the engine with JUnit 5 unit tests and MockMvc integration tests, and validated request payloads with Jakarta Bean Validation.

## CERTIFICATIONS

**Java Programming: Solving Problems with Software** – Duke University, Coursera

**Data Structures** – University of California San Diego, Coursera

**Advanced Java** – LearnKartS, Coursera

**Neural Networks and Deep Learning** – DeepLearning.AI, Coursera